

Name: _____

Class: _____

ACTIVITY SHEET

Chapter 1: Revision

1 Match each term with the appropriate description using an arrow.

a atoms
b nucleus
c proton
d neutron
e electron
f electron cloud
g energy level
h element
i chemical symbol
j atomic number
k mass number
l electrostatic attraction
m nuclear force
n isotopes
o radioactive decay
p relative atomic mass
q periodic table
r allotropes
s electron configuration
t light spectrum
u absorption spectroscopy
v mass spectrometry

A number of protons in the nucleus of an atom
B arrangement of electrons around the nucleus in energy levels
C positively charged sub-atomic particle found in the nucleus
D chart of elements arranged according to increasing atomic number
E force that binds neutrons and protons together in the nucleus
F region around nucleus containing electrons with equal energy
G particles of which all matter is made
H spontaneous disintegration of unstable atoms
I dense central region of an atom
J substance made of atoms with the same number of protons
K number of protons and neutrons in the nucleus of an atom
L technique to determine the concentration of an element
M attraction between oppositely charged objects
N atoms of an element with varying numbers of neutrons
O different physical forms of the same element
P each element has a unique one
Q uncharged sub-atomic particle found in the nucleus
R different wavelengths of light admitted by an excited atom
S technique using particle masses to measure relative abundance
T mean mass of an element taking into account isotope abundance
U negatively charged sub-atomic particle found outside the nucleus
V region about nucleus where electrons are likely to be found

2 Explain the differences between the terms in each of the items below.

a atomic number and mass number

b isotopes and allotropes of an element

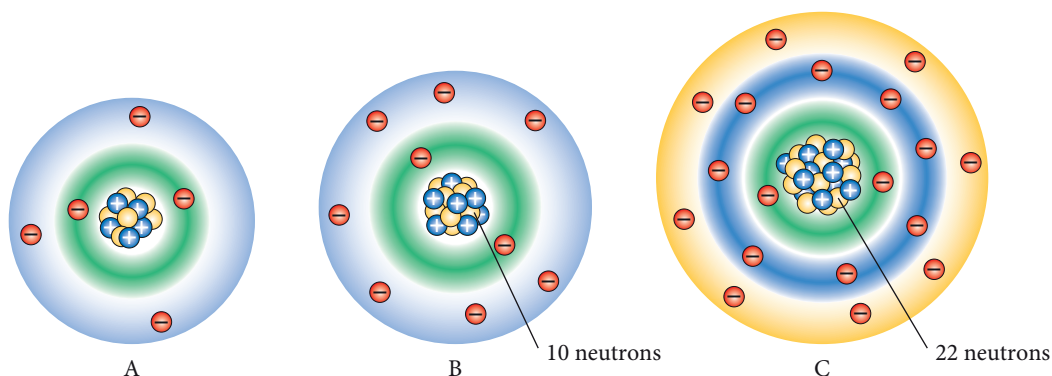
c a period and a group of the periodic table

d atomic absorption spectroscopy and mass spectrometry

3 Complete the table below, comparing the properties of subatomic particles.

Particle	Symbol	Charge	Location	Relative mass
Proton				
Electron				
Neutron				

4 Study each of the three electrically neutral atoms below. Then complete the table that follows.

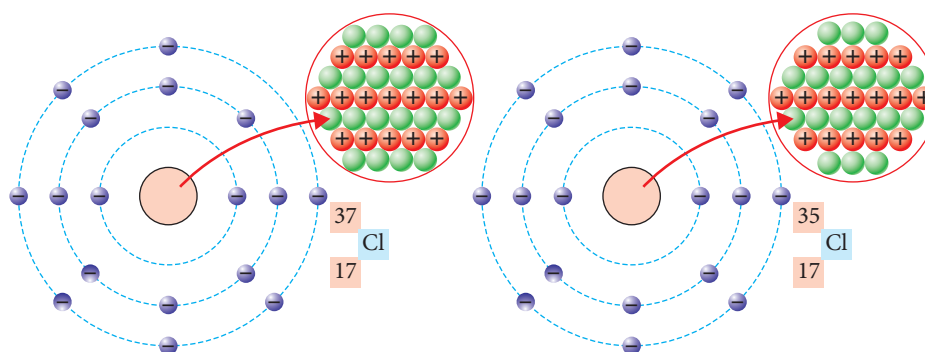


	Atom A	Atom B	Atom C
Number of protons			
Number of electrons			
Number of neutrons (N)			
Atomic number (A)			
Mass number (Z)			
Element symbol			
Atomic representation (A_ZX)			

5 Identify the force that holds an atom together and the force that holds its nucleus together.

6 Explain why most non-naturally occurring elements rapidly break down into other elements.

7 The diagram shows the structure of two naturally occurring isotopes of the element chlorine.



a Determine how many protons there are in an atom of chlorine.

b Calculate the number of neutrons in an atom of each isotope above using the formula:

$$A = Z + N$$

c Explain why both atomic structures represent isotopes of chlorine.

The percentages of the two isotopes in a sample of chlorine gas are: 76% ^{35}Cl , 24% ^{37}Cl .

d Find the relative atomic mass of the sample of chlorine.

8 Electrons are found in specific energy levels (energy shells) around the nucleus of the atom.

a State how many electrons can be accommodated by the first, second and third shells.

b Describe the trends for the electronegativities of atoms.

c Describe the trends for the ionisation energies of atoms.

d Describe the trends for electron affinities of atoms.

10 The left diagram shows the emission spectrum of hydrogen atoms and the right diagram shows the emission spectrum of lithium atoms.



a Describe the difference between the ground and excited states of an atom.

b Explain what happens when an electron absorbs an appropriate amount of energy.

c What do the brightly coloured lines on the spectra above represent?

d Why are the spectra of the two elements different?